COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:		**DATE SUBMITTED:**	
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SEATWORK 9.1 Manual Installation Procedure for Prometheus

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Prepare the Server
List the commands required to:
1. Update the package repository.
2. Upgrade installed packages.
3. Install the utilities required by the installation.

```
tadalan-workdesk@workstation:~$ sudo apt update

Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,283 kB]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 http://ph.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,697 kB]
Fetched 3,106 kB in 2s (1,802 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
208 packages can be upgraded. Run 'apt list --upgradable' to see them.
tadalan-workdesk@workstation:~$ sudo apt install -y wget tar
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.5).
wget set to manually installed.
tar is already the newest version (1.35+dfsg-3ubuntu0.4).
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 208 not upgraded.
```

Task 2: Create the Prometheus User
Identify the commands required to:
1. Create a dedicated prometheus system user.
2. Prevent the user from having a normal login shell.
3. Verify that the user was successfully created.

```
tadalan-workdesk@workstation:~$ sudo useradd --no-create-home --system prometheus
tadalan-workdesk@workstation:~$ sudo usermod --shell /usr/sbin/nologin prometheus
tadalan-workdesk@workstation:~$ id prometheus
uid=997(prometheus) gid=984(prometheus) groups=984(prometheus)
tadalan-workdesk@workstation:~$
```

Part 3: Create Prometheus Directories
Identify the commands required to:
1. Create /etc/prometheus.
2. Create /var/lib/prometheus.
3. Assign the appropriate owner and group.
4. Verify the directory permissions and ownership.

```
tadalan-workdesk@workstation:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
tadalan-workdesk@workstation:~$ sudo mkdir -p /etc/prometheus
tadalan-workdesk@workstation:~$ sudo mkdir -p /var/lib/prometheus
tadalan-workdesk@workstation:~$ sudo chown prometheus:prometheus /etc/prometheus
tadalan-workdesk@workstation:~$ sudo chown prometheus:prometheus /var/lib/prometheus
tadalan-workdesk@workstation:~$ ls -ld  /etc/prometheus
drwxr-xr-x 2 prometheus prometheus 4096 Sep 22 09:26 /etc/prometheus
tadalan-workdesk@workstation:~$ ls -ld  /var/lib/prometheus
drwxr-xr-x 2 prometheus prometheus 4096 Sep 22 09:27 /var/lib/prometheus
tadalan-workdesk@workstation:~$
```

Part 4: Download Prometheus
Identify the commands required to:
1. Navigate to a temporary working directory.
2. Download the Prometheus archive.
3. Verify that the downloaded file exists.
4. Extract the Prometheus archive.
5. Verify the extracted directory and its contents.
Include the Prometheus version used in your procedure.

```
tadalan-workdesk@workstation:/tmp$ wget https://github.com/prometheus/prometheus
/releases/download/v3.14.0/prometheus-3.14.0.linux-amd64.tar.gz
--2026-09-22 09:38:07--  https://github.com/prometheus/prometheus/releases/downl
oad/v3.14.0/prometheus-3.14.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.205.243.166
Connecting to github.com (github.com)|20.205.243.166|:443... connected.
HTTP request sent, awaiting response... 302 Found
```

```
tadalan-workdesk@workstation:/tmp$ ls -lh prometheus-3.14.0.linux-amd64.tar.gz
-rw-rw-r-- 1 tadalan-workdesk tadalan-workdesk 103M Aug 18 15:36 prometheus-3.14
.0.linux-amd64.tar.gz
tadalan-workdesk@workstation:/tmp$ tar -xvf prometheus-3.14.0.linux-amd64.tar.gz
prometheus-3.14.0.linux-amd64/
prometheus-3.14.0.linux-amd64/prometheus.yml
prometheus-3.14.0.linux-amd64/prometheus
prometheus-3.14.0.linux-amd64/NOTICE
prometheus-3.14.0.linux-amd64/LICENSE
prometheus-3.14.0.linux-amd64/promtool
tadalan-workdesk@workstation:/tmp$ ls =lh
ls: cannot access '=lh': No such file or directory
tadalan-workdesk@workstation:/tmp$ ls -lh
total 103M
drwxr-xr-x 2 tadalan-workdesk tadalan-workdesk 4.0K Aug 18 15:31 prometheus-3.14
.0.linux-amd64
-rw-rw-r-- 1 tadalan-workdesk tadalan-workdesk 103M Aug 18 15:36 prometheus-3.14
.0.linux-amd64.tar.gz
drwx------ 6 root             root             4.0K Sep 22 09:36 snap-private-tm
p
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-colord.service-arI5mP
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-colord.service-arI5mP
drwx------ 3 root             root             4.0K Sep 22 08:44 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-fwupd.service-2edYsW
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-ModemManager.service-fT2MZ5
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-polkit.service-MxYoTK
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-power-profiles-daemon.service-Hwv9KN
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-switcheroo-control.service-vOLAtV
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-systemd-logind.service-Ywtg2Y
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-systemd-oomd.service-olSTnt
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-systemd-resolved.service-FgOKhp
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-systemd-timesyncd.service-tFs0QV
drwx------ 3 root             root             4.0K Sep 22 08:38 systemd-private
-5dcbfc47773f47cfb77cfb289c2addcf-upower.service-UPT7R7
```

```
tadalan-workdesk@workstation:/tmp$ ls -lh prometheus-3.14.0.linux-amd64/
total 247M
-rw-r--r-- 1 tadalan-workdesk tadalan-workdesk  12K Aug 18 15:31 LICENSE
-rw-r--r-- 1 tadalan-workdesk tadalan-workdesk 4.3K Aug 18 15:31 NOTICE
-rwxr-xr-x 1 tadalan-workdesk tadalan-workdesk 136M Aug 18 15:31 prometheus
-rw-r--r-- 1 tadalan-workdesk tadalan-workdesk 1.2K Aug 18 15:31 prometheus.yml
-rwxr-xr-x 1 tadalan-workdesk tadalan-workdesk 112M Aug 18 15:31 promtool
tadalan-workdesk@workstation:/tmp$
```

Task 5: Install the Prometheus Binaries
Identify the commands required to:
1. Locate the prometheus binary.
2. Locate the promtool binary.
3. Copy both binaries to /usr/local/bin.
4. Set the appropriate ownership.
5. Verify the installed Prometheus version.
6. Verify the installed promtool version.

```
tadalan-workdesk@workstation:/tmp$ find /tmp/prometheus-3.14.0.linux-amd64 -name prome
theus
/tmp/prometheus-3.14.0.linux-amd64/prometheus
tadalan-workdesk@workstation:/tmp$ find /tmp/prometheus-3.14.0.linux-amd64 -name promt
ool
/tmp/prometheus-3.14.0.linux-amd64/promtool
```

```
tadalan-workdesk@workstation:/tmp$ sudo cp /tmp/prometheus-3.14.0.linux-amd64/promethe
us /usr/local/bin
tadalan-workdesk@workstation:/tmp$ sudo cp /tmp/prometheus-3.14.0.linux-amd64/promtool
/usr/local/bin
```

```
tadalan-workdesk@workstation:/tmp$ sudo cp /tmp/prometheus-3.14.0.linux-amd64/promtool
 /usr/local/bin
tadalan-workdesk@workstation:/tmp$ sudo chown prometheus:prometheus /usr/local/bin/pro
metheus
tadalan-workdesk@workstation:/tmp$ sudo chown prometheus:prometheus /usr/local/bin/pro
mtool
tadalan-workdesk@workstation:/tmp$ prometheus --version
prometheus, version 3.14.0 (branch: HEAD, revision: d7598b7141418fa35be2b5ec5d0fefb634
199610)
  build user:       root@f423027f4410
  build date:       20260817-16:49:19
  go version:       go1.26.6
  platform:         linux/amd64
  tags:             netgo,builtinassets
tadalan-workdesk@workstation:/tmp$ promtool --version
promtool, version 3.14.0 (branch: HEAD, revision: d7598b7141418fa35be2b5ec5d0fefb63419
9610)
  build user:       root@f423027f4410
  build date:       20260817-16:49:19
  go version:       go1.26.6
  platform:         linux/amd64
  tags:             netgo,builtinassets
tadalan-workdesk@workstation:/tmp$
```

Task 6: Install the Prometheus Configuration Files
Identify the commands required to:
1. Copy prometheus.yml to /etc/prometheus.
2. Copy the consoles directory.
3. Copy the console_libraries directory.
4. Set the appropriate ownership.
5. Verify the files and directories.



Task 7: Validate the Configuration

1. Identify the command used to check whether the Prometheus configuration is valid.
2. Explain why configuration validation should be performed before starting the Prometheus service.
    - *This validates the Prometheus configuration file fist before starting the service.*

```
root@workstation:/tmp# promtool check config /etc/prometheu
s/prometheus.yml
Checking /etc/prometheus/prometheus.yml
 SUCCESS: /etc/prometheus/prometheus.yml is valid prometheu
s config file syntax
```

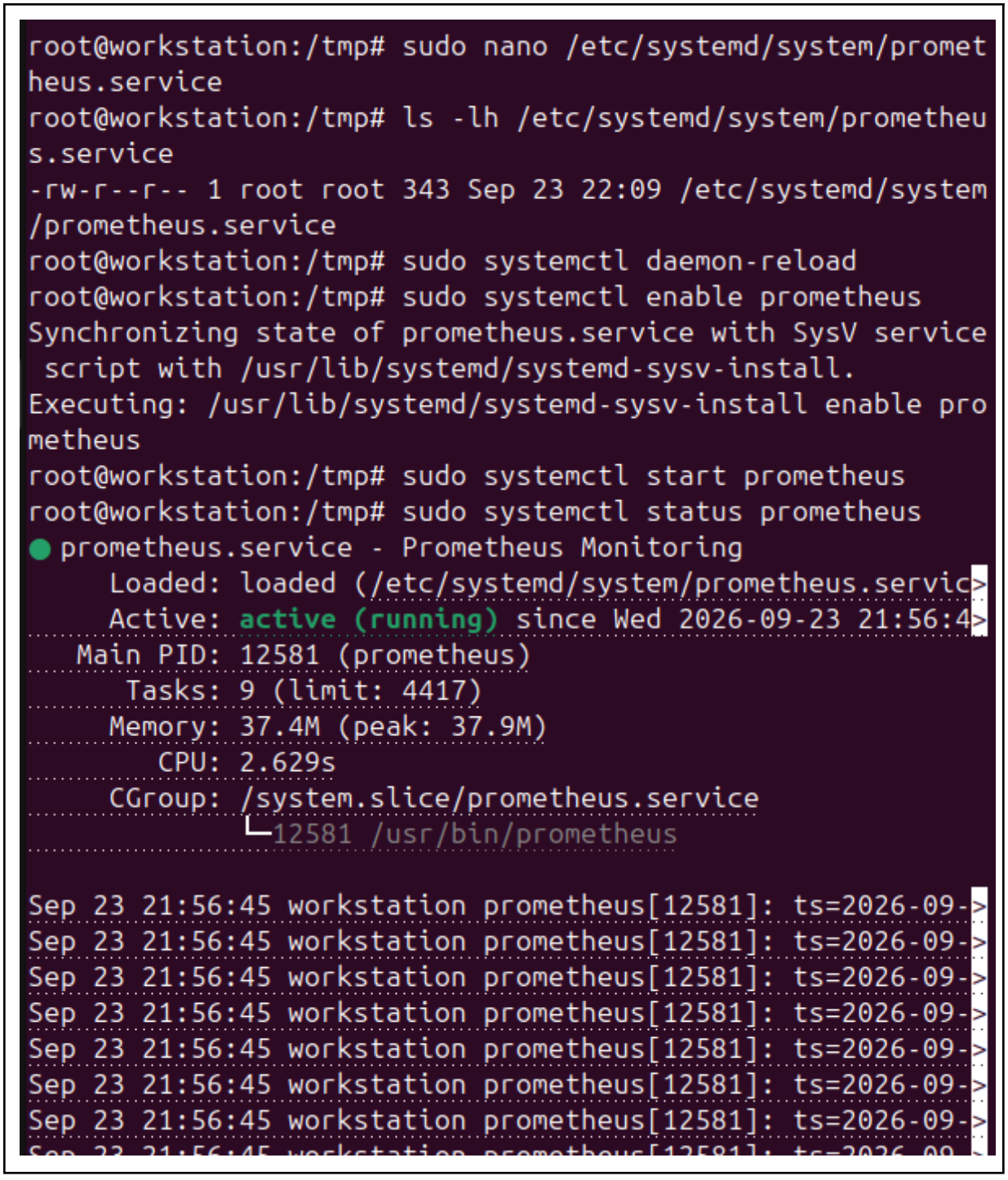
Task 8: Create the systemd Service
Identify the procedure for creating: /etc/systemd/system/prometheus.service Your documentation must include the complete service file.
Explain the purpose of the important directives, including:

- User
- Group
- ExecStart
- --config.file
- --storage.tsdb.path
- Restart
- WantedBy

- *The service file tells Linux how to start and manage Prometheus automatically.*

```
root@workstation:/tmp# sudo nano /etc/systemd/system/promet
heus.service
root@workstation:/tmp#
```

Task 9: Start and Enable Prometheus
Identify the commands required to:
1. Reload systemd.
2. Enable Prometheus to start automatically during boot.
3. Start Prometheus.
4. Check the service status.



Task 10: Verify Prometheus

Identify the commands required to verify that Prometheus is running correctly.

Your procedure must include:

1. Checking whether port 9090 is listening.

2. Testing Prometheus locally.
3. Checking the Prometheus readiness endpoint.
4. Checking the Prometheus version.
5. Checking the systemd service status.

```
Get:9 http://ph.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 840 kB in 1s (876 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
38 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@workstation:/tmp# apt install -y curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 38 not upgraded.
Need to get 226 kB of archives.
After this operation, 536 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.13 [226 kB]
Fetched 226 kB in 0s (1,071 kB/s)
Selecting previously unselected package curl.
(Reading database ... 237465 files and directories currentl
y installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.13_amd64.deb .
..
Unpacking curl (8.5.0-2ubuntu10.13) ...
Setting up curl (8.5.0-2ubuntu10.13) ...
Processing triggers for man-db (2.12.0-4build2) ...
root@workstation:/tmp# curl http://localhost:9090
<a href="/classic/graph">Found</a>.
```

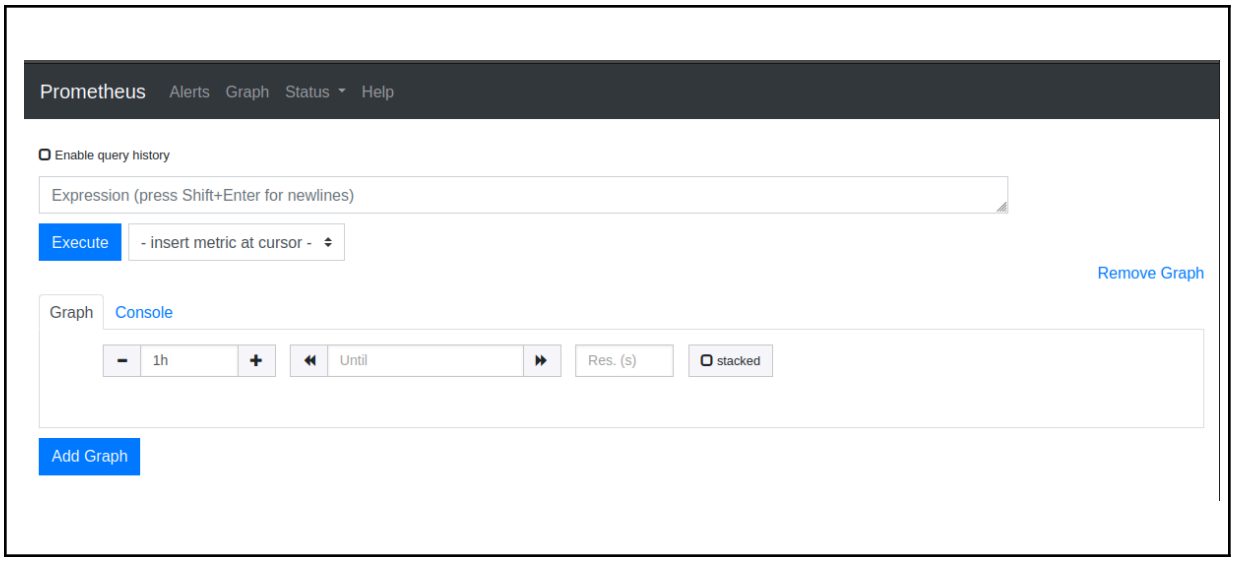
```
root@workstation:/tmp# curl http://localhost:9090
<a href="/classic/graph">Found</a>.

root@workstation:/tmp# curl http://localhost:9090/-/ready
Prometheus Server is Ready.
root@workstation:/tmp# prometheus --version
prometheus, version 3.14.0 (branch: HEAD, revision: d7598b7141418fa35be2b5ec5d0fefb634199610)
  build user:       root@f423027f4410
  build date:       20260817-16:49:19
  go version:       go1.26.6
  platform:         linux/amd64
  tags:             netgo,builtinassets
root@workstation:/tmp# apt install -y curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.13).
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 38 not upgraded.
root@workstation:/tmp#
```

Part 11: Access Prometheus Through a Browser

Identify how Prometheus can be accessed from another computer.
Document the URL format: http://<SERVER-IP>:9090
Explain:
1. What <SERVER-IP> represents.
2. Why port 9090 is used.
3. What should appear when Prometheus is successfully accessed.

- The server IP identifies the computer running Prometheus firs, at the same time the port 9090 is used in order to access its web interface. As demonstrated it was successful, as the Prometheus web interface did appear in the browser.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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